Project Initialization and Planning Phase

Date	5 July 2026
Team ID	
Project Title	A Comprehensive Measure of Well-Being
Maximum Marks	5 Marks

Project Proposal (Proposed Solution)

The **Human Development Index (HDI) Prediction System** is designed to predict the Human Development Index (HDI) category of a country using Machine Learning techniques. HDI is an internationally recognized indicator that measures the overall development of a country based on **Life Expectancy at Birth, Expected Years of Schooling, Mean Years of Schooling, and Gross National Income (GNI) per Capita**.

The proposed solution uses historical Human Development datasets along with a **Random Forest Classifier** to analyze socio-economic indicators and accurately classify countries into **Very High, High, Medium, or Low Human Development** categories.

The system is implemented as a **Flask-based web application**, allowing users to enter development indicators through an interactive interface and instantly receive the predicted HDI category. This solution assists policymakers, researchers, students, and analysts in understanding development trends and supporting data-driven decision-making.

Project Overview

Objective	The primary objective of this project is to develop a Machine Learning model capable of accurately predicting the **Human Development Index (HDI) category** using socio-economic indicators such as Life Expectancy at Birth, Expected Years of Schooling, Mean Years of Schooling, and Gross National Income (GNI) per Capita. The system aims to assist governments, researchers, students, and policymakers in understanding development trends and supporting evidence-based decision-making.
Scope	The project focuses on: • Collecting and understanding Human Development Index datasets. • Data preprocessing and cleaning. • Feature selection. • Training a Random Forest Classification model. • Evaluating prediction performance. • Deploying the trained model using Flask. • Providing an interactive web application for instant HDI prediction. The solution supports educational research, policy analysis, and socio-economic development planning.
Problem Statement	
Description	Predicting the Human Development Index manually is difficult because it depends on multiple socio-economic indicators that influence one another. Traditional analysis is time-consuming and often fails to identify hidden relationships within large datasets.
Impact	Inaccurate or delayed HDI prediction may affect government planning, policy formulation, educational development, healthcare improvement, and economic growth strategies. A Machine Learning-based prediction system provides faster and more reliable predictions, enabling informed decisions and efficient resource allocation.

Proposed Solution	
Approach	Develop a Machine Learning-based prediction system using the **Random Forest Classifier** algorithm. The model analyzes the following indicators: • Life Expectancy at Birth • Expected Years of Schooling • Mean Years of Schooling • Gross National Income (GNI) per Capita The trained model predicts the Human Development Index category and displays the result through a Flask web application.
Key Features	• Dataset preprocessing and cleaning • Handling missing values • Feature selection • Label Encoding • Random Forest Classification • HDI Category Prediction • Model Saving using Pickle (.pkl) • Interactive Flask Web Application • User-friendly interface with responsive design • Instant prediction results

Resource Requirements

Resource Type	Description	Specification/Allocation
Hardware		
Processor	Computing Resources	Intel Core i5 or higher

Memory	RAM specifications	8 GB
Storage	Disk space	20 GB Free SSD/HDD
Software		
Operating System	Platform	Windows 10/11 or Ubuntu
Programming Language	Development	Python 3.10+
Framework	Backend	Flask
Libraries	Machine Learning	Pandas, NumPy, Scikitlearn, Matplotlib, Joblib
IDE	Development	VS Code
Development Environment	Notebook	Jupyter Notebook
Data		

Data	Source, size, format	Human Development Index Dataset (CSV) containing socioeconomic indicators collected from public development datasets. Source: Kaggle Human Development Index Dataset